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1 Introduction to Python

Python is a high-level interpreted, open-source programming language that is widely used for software development, automation, data analysis, artificial intelligence, machine learning, web development, scientific computing, and many other domains.

2 Roles in Analytics

According to the uploaded material, common analytics role include:

- Data Engineer
- Data Analyst/ Business Analyst
- Data Scientist/ Decision Scientist
- Machine Learning/ Artificial Intelligence Engineer
- MLOps (Machine Learning Operations)

3 Types of Analytics

- Descriptive Analytics
- Diagnostic Analytics
- Inferential Analytics
- Predictive Analytics
- Prescriptive Analytics
- Cognitive Analytics

4 Jupyter Notebook

Jupyter Notebook is an interactive development environment that allows users to combine executable Python code, formatted text, mathematical equations, tables, and visualizations in a single document.

4.1 Jupyter Notebook State

Every notebook cell has two states.

4.1.1 Edit State

Used for editing the contents of a cell.

Common shortcuts:

- Ctrl+Enter: Execute current cell and remain in the same cell.
- Shift+Enter: Execute current cell and move to the next cell.

4.1.2 Select State

Used for notebook management.

Important shortcuts:

- ESC+C: Copy cell
- ESC+X: Cut cell
- ESC+V: Paste cell

- ESC+A: Insert cell above
- ESC+B: Insert cell below
- ESC+DD: Delete current cell

4.2 Jupyter Notebook Modes

Three notebook modes are introduced.

- **Code Mode:** Used for writing executable Python code.
Shortcut: ESC+Y
- **RawNBConvert Mode:** Behaves like plain text without formatting.
Shortcut: ESC+R
- **Markdown Mode:** Used to create documentation.
Shortcut: ESC+M

5 Comments

Comments are not executed by the Python interpreter. Primary uses include:

- Documentation
- Recording observations.

5.1 Single-Line Comment

Begins with the symbol #

5.2 Multi-Line String

Python does not have true multi-line comments.

Instead, triple-quoted strings are commonly used for documentation (docstrings).

The uploaded code is reproduced exactly below.

```
# This is the comment
```

```
""" This is
multi-line string"""
```